



LEEDS CITY COUNCIL City Development

Selectapost 6
The Leonardo Building
2 Rossington Street
Leeds LS2 8HB

Switchboard: (0113) 2348080
Telex: (0113) 2556170
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Junction	Rodley Roundabout, A6120 Ring Rd – Rodley Lane – Calverley Lane		
Company	Telent	Serial No	New / Modification
Mains Supply:-	230V/50Hz	High Intensity:- YES	Dimming:- 32V
Engineer	A Bradley	Telephone N° 0113 2476785	

NOTE TO UTC PERSONNEL

This specification should be used for general purposes only.
The Signal Company documentation should be used for controller operation
and timings unless amended on this sheet as RAM data.

RAM UPDATE (if no data then write NONE)	Reason for change	Initials	Date
Write date current EPROM fitted:-			

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337L - Rodley Roundabout, A6120 Ring Rd – Rodley Lane – Calverley Lane

A Telent Optima 32-phase ELV controller from contract 500/3435 shall provide the method of control in this specification

Mode of Operation

This site shall run UTC / CLF mode with Fixed Time mode fall back, initially set to mimic CLF (day time) plan 1.
VA mode not configured.

Communication

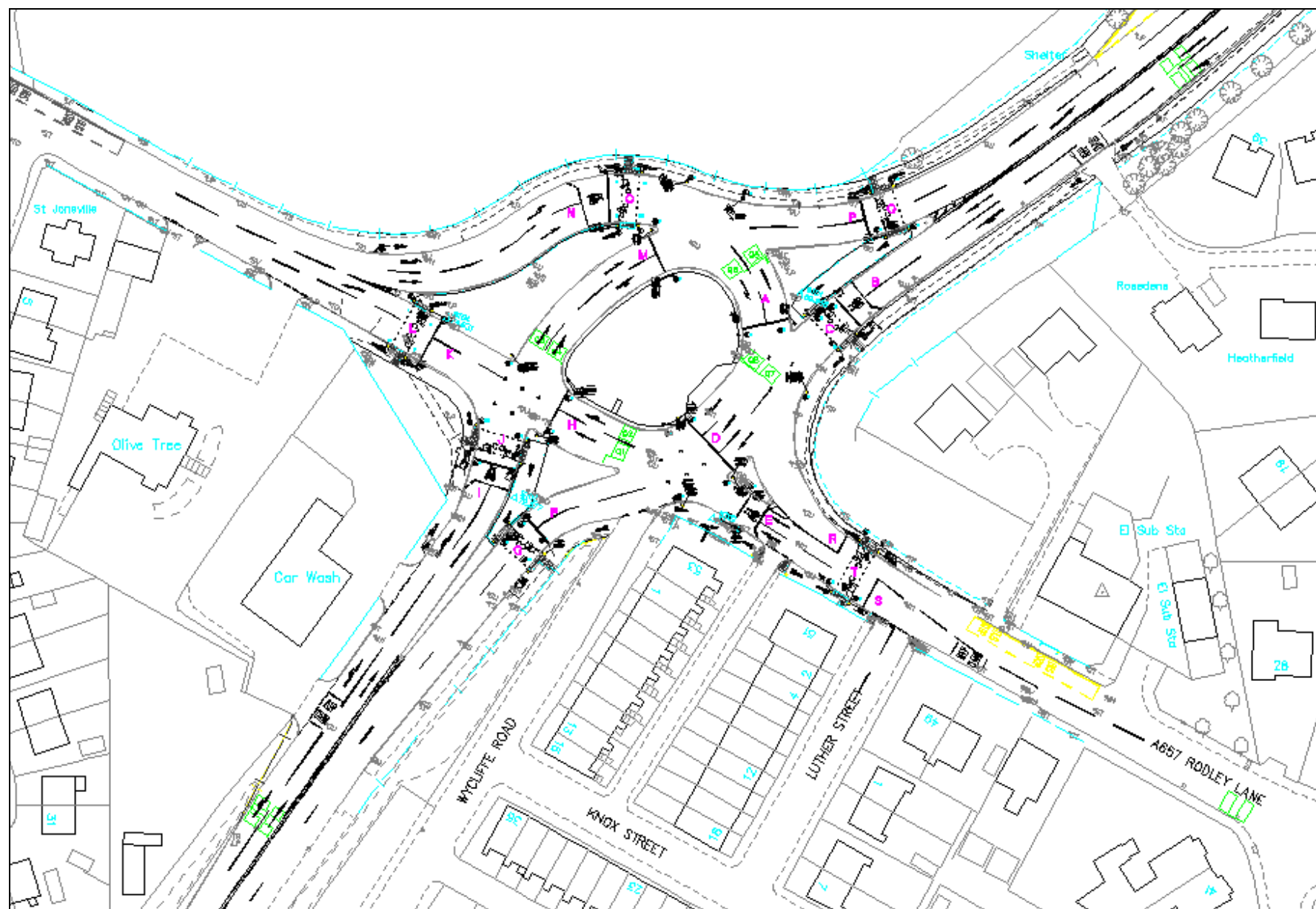
A 64-bit Imtech Chameleon shall be used for monitoring and control purposes.

UTC Mode - Cross Linking

The controller shall be configured so that ALL STREAMS must run UTC simultaneously. Should UTC release on any stream, ALL OTHER STREAMS shall also drop to the next available mode of control.

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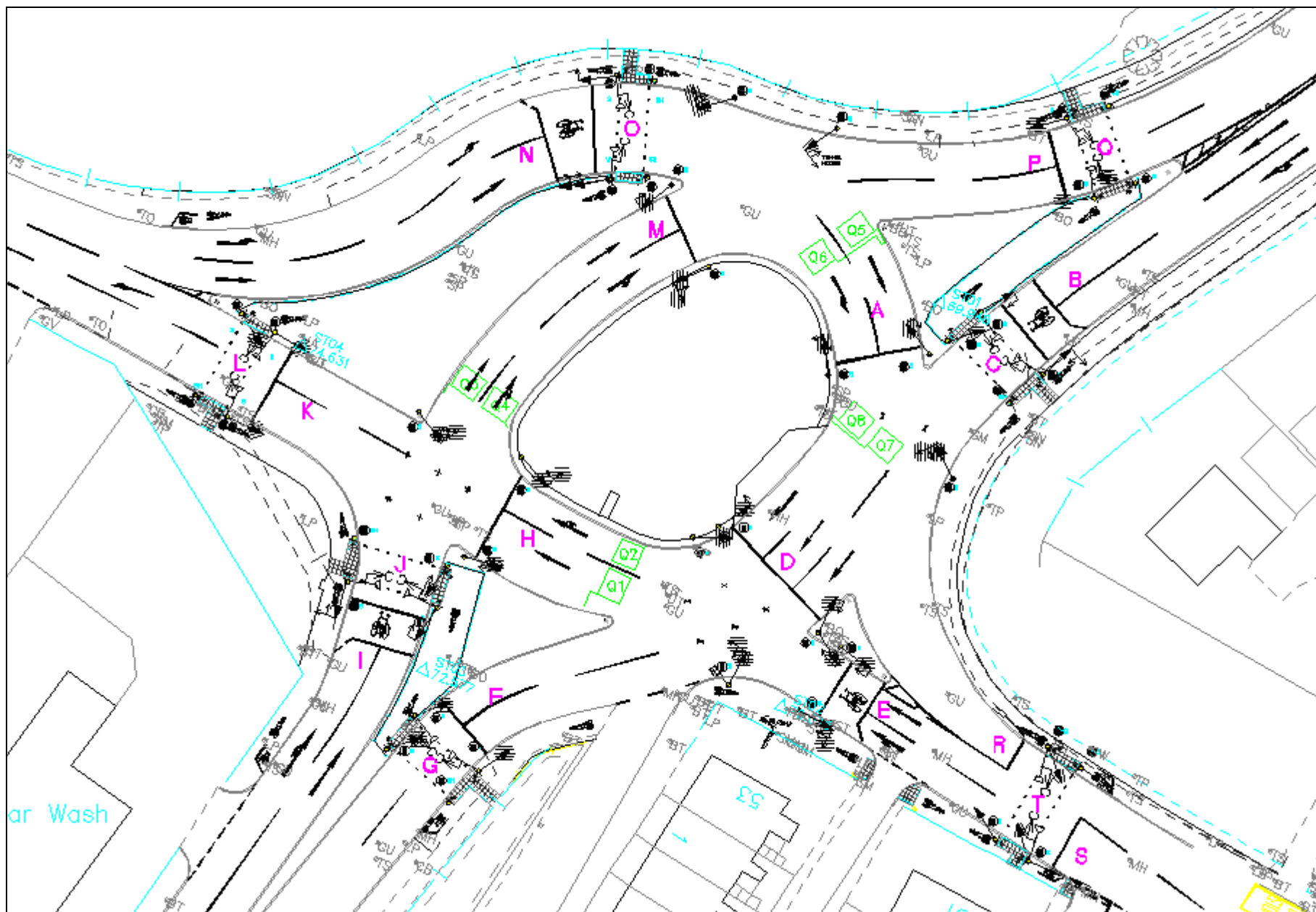
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NOT TO SCALE

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NOT TO SCALE

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BASIC SITE DATA FOR STREAM N° 1 & 2

Please sketch stage movements and annotate phases

Stage (1)	Stage (2)			Stage (3) ALL RED STREAM 1 DUMMY U
Stage (4)	Stage (5)			Stage (6) ALL RED STREAM 2 DUMMY V

	STREAM 1	STREAM 2		
ALL RED STAGE	1	4		
START UP STAGE	1	4		

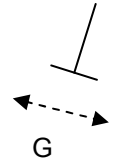
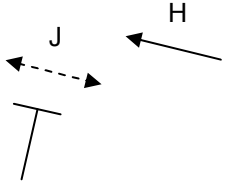
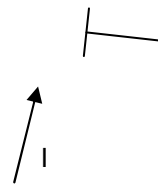
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BASIC SITE DATA FOR STREAM N° 3 & 4

Please sketch stage movements and annotate phases

Stage (7) 	Stage (8) 			Stage (9) ALL RED STREAM 3 DUMMY W
Stage (10) 	Stage (11) 			Stage (12) ALL RED STREAM 4 DUMMY X

	STREAM 3	STREAM 4		
ALL RED STAGE	7	10		
START UP STAGE	7	10		

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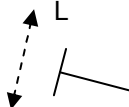
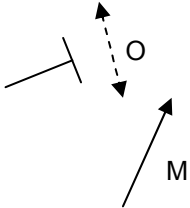
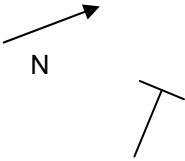
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BASIC SITE DATA FOR STREAM N° 5 & 6

Please sketch stage movements and annotate phases

Stage (13) 	Stage (14) 			Stage (15) ALL RED STREAM 5 DUMMY Y
Stage (16) 	Stage (17) 			Stage (18) ALL RED STREAM 6 DUMMY Z

	STREAM 5	STREAM 6		
ALL RED STAGE	13	16		
START UP STAGE	13	16		

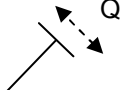
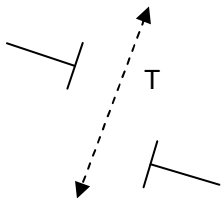
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BASIC SITE DATA FOR STREAM N° 7 & 8

Please sketch stage movements and annotate phases

Stage (19) 	Stage (20) 			Stage (21) ALL RED STREAM 7 DUMMY A1
Stage (22)  	Stage (23) 			Stage (24) ALL RED STREAM 8 DUMMY A2

	STREAM 7	STREAM 8		
ALL RED STAGE	19	22		
START UP STAGE	19	22		

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USE OF PHASES LOCATION		PHASE TYPE NOTE 3	APPEAR TYPE NOTE 1	TERMINATION		PED BLKT	MIN GREEN	MAXIMUMS			
								(Day)	(AM)	(PM)	(Night)
				TYPE NOTE 2	ASSOC PHASE			SET A	SET B	SET C	SET D
A	Eastbound internal	T	0	0	-	-	7				
B	Ring Road Southbound	T	0	0	-	-	7				
C	Ring Road Southbound Ped	P	0	0		3	5				
D	Southbound internal	T	0	0	-	-	7				
E	Rodley Lane	T	0	0	-	-	7				
F	Far Ring Road Southbound	T	0	0	-	-	7				
G	Far Ring Road Southbound Ped	P	0	0		3	5				
H	Westbound internal	T	0	0	-	-	7				
I	Ring Road Northbound	T	0	0	-	-	7				
J	Ring Road Northbound Ped	P	0	0		3	5				
K	Calverley Lane Westbound	T	0	0	-	-	7				
L	Calverley Lane Westbound Ped	P	0	0		3	5				
M	Northbound internal	T	0	0	-	-	7				
N	Calverley Lane	T	0	0	-	-	7				
O	Calverley Lane Ped	P	0	0		3	6				
P	Far Ring Road Northbound	T	0	0	-	-	7				
Q	Far Ring Road Northbound Ped	P	0	0		3	5				
R	Rodley Lane Westbound	T	0	0	-	-	7				
S	Rodley Lane Eastbound	T	0	0	-	-	7				
T	Rodley Lane Ped	P	0	0		3	6				
U	Dummy U Str.1	D	0	0			3				
V	Dummy V Str.2	D	0	0			2				
W	Dummy W Str.3	D	0	0			3				
X	Dummy X Str.4	D	0	0			4				
Y	Dummy Y Str.5	D	0	0			4				
Z	Dummy Z Str.6	D	0	0			5				
A1	Dummy A1 Str.7	D	0	0			2				
A2	Dummy A2 Str.8	D	0	0			5				

Note 1:- Condition of Appearance

0 - Phase always appears when stage runs.

1 - Phase only appears if demand exists at start of inter-stage

2 - Phase appears, if demanded, at any time up until the end of the stage in which the phase runs

Note 2:- Condition of Termination

0 - Terminates at end of stage

1 - Terminates when associated phase gains right of way

2 - Terminates when associated phase loses right of way

Note 3:- Phase Types

T - Traffic

P - Pedestrian

F - Filter Arrow

I - Indicative Arrow

L - L.R.T.

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3 - Phase appears, if demanded at any time during the stage up until the window time expires

S - Switched Sign
D - Dummy

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PHASES		CONFLICTING/OPPOSING PHASES																													
		A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A1	A2		
A			C	.																			C								
B		C		C																			C								
C		.	C																				C								
D																															
E																								C							
F																								C							
G																															
H																															
I																															
J																															
K																															
L																															
M																															
N																															
O																															
P																															
Q																															
R																															
S																															
T																															
U																															
V																															
W																															
X																															
Y																															
Z																															
A1																															
A2																															

C = CONFLICTING &
OPPOSING

O = OPPOSING ONLY

. = NO CONFLICT

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				USE OF STAGES - STAGE/PHASE RELATIONSHIP																									
STAGE	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A1	A2	
1	✓		✓																										
2		✓																											
3	A	L	L		R	E	D		S	T	R	E	A	M		1					✓								
4				✓																									
5					✓																								
6	A	L	L		R	E	D		S	T	R	E	A	M		2						✓							
7						✓																							
8							✓																						
9	A	L	L		R	E	D		S	T	R	E	A	M		3							✓						
10								✓		✓																			
11									✓																				
12	A	L	L		R	E	D		S	T	R	E	A	M		4								✓					
13											✓																		
14												✓																	
15	A	L	L		R	E	D		S	T	R	E	A	M		5									✓				
16													✓		✓														
17														✓															
18	A	L	L		R	E	D		S	T	R	E	A	M		6										✓			
19																✓													
20																	✓												
21	A	L	L		R	E	D		S	T	R	E	A	M		7											✓		
22																		✓	✓										
23																				✓									
24	A	L	L		R	E	D		S	T	R	E	A	M		8												✓	

STREAM	WINDOW TIME
1	
1	
1	
2	
2	
2	
3	
3	
3	
4	
4	
4	
5	
5	
5	
6	
6	
6	
7	
7	
7	
8	
8	
8	

MAX REVERTIVE DEMAND PHASE																												
				ARTERIAL REVERSION STAGE OR PHASE PER STREAM FOR REVERSION IN THE ABSENCE OF DEMANDS - N/A																								

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PHASE	PHASE INTERGREEN TIMES																											
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A1	A2
A		5	.																		3							
B	6		6																		3							
C	.	8																			3							
D					5																	3						
E				7																		3						
F							5																3					
G						8																	3					
H									5	.														3				
I								6		6														1				
J								.	9															3				
K												5													0			
L											9														3			
M													5	.												3		
N												6		6												0		
O												.	10													3		
P																	5										3	
Q																8											0	
R																			.	5								3
S																		.		5								3
T																		10	10									0
U	2	2	0																									
V				2	2																							
W						2	0																					
X								2	2	0																		
Y											2	0																
Z													2	2	0													
A1																2	0											
A2																		2	2	0								

STARTING	
INTERGREEN	
PERIOD	10

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ADDITIONAL PHASE DELAYS

N°	Delay Phase	Change From	To Stage	By (X) Secs	N°	Delay Phase	Change From	To Stage	By (X) Secs	N°	Delay Phase	Change From	To Stage	By (X) Secs
1	A	1	2	3	31					61				
2	H	10	11	4	32					62				
3	M	16	17	5	33					63				
4					34					64				
5					35					65				
6					36					66				
7					37					67				
8					38					68				
9					39					69				
10					40					70				
11					41					71				
12					42					72				
13					43					73				
14					44					74				
15					45					75				
16					46					76				
17					47					77				
18					48					78				
19					49					79				
20					50					80				
21					51					81				
22					52					82				
23					53					83				
24					54					84				
25					55					85				
26					56					86				
27					57					87				
28					58					88				
29					59					89				
30					60					90				

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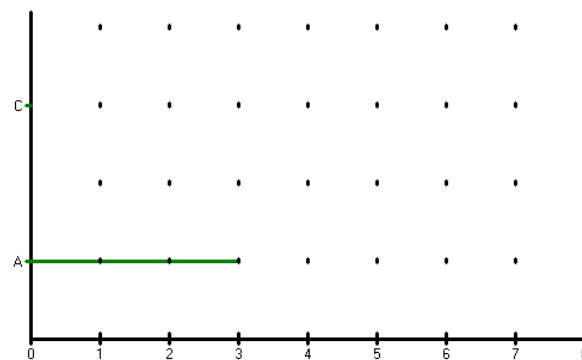
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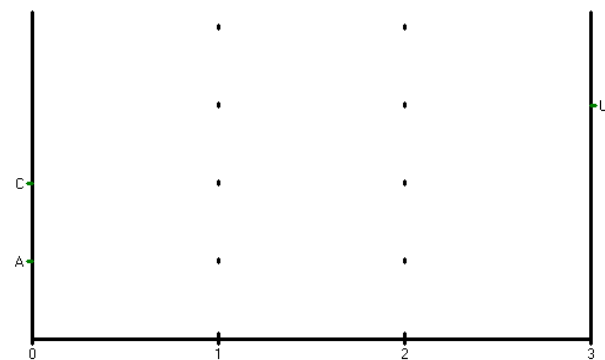
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337L - Rodley – Ring Road Roundabout – Stream 1

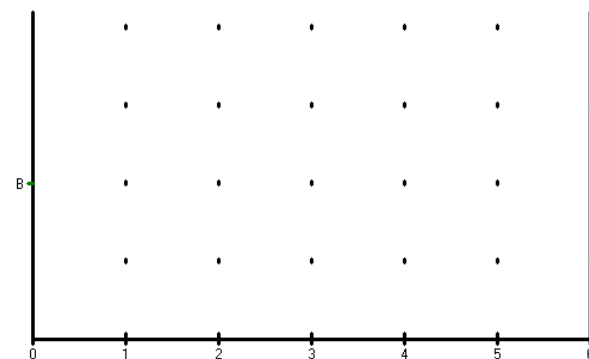
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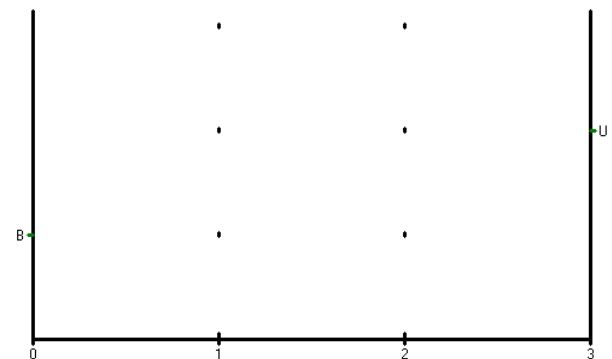
Change From 1 To 3



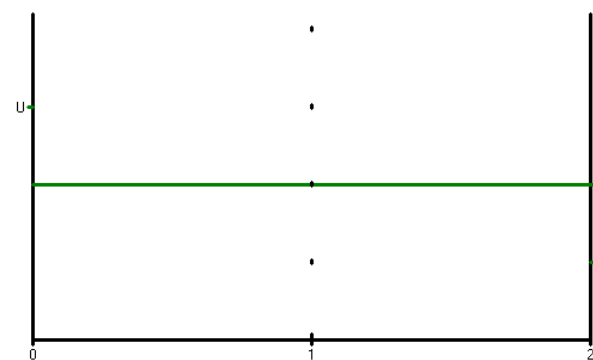
Change From 2 To 1



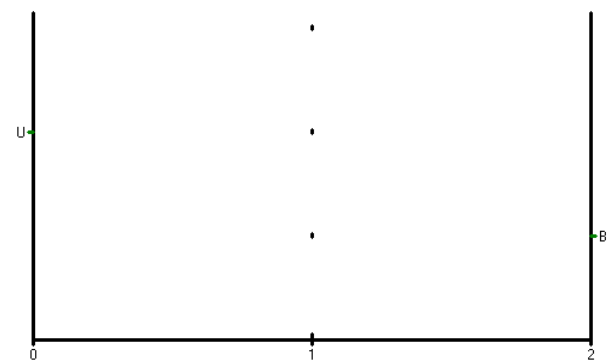
Change From 2 To 3



Change From 3 To 1



Change From 3 To 2



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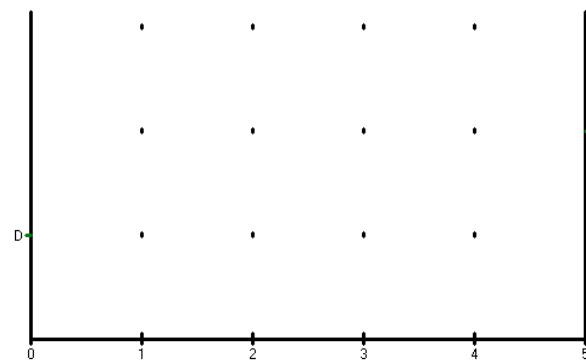
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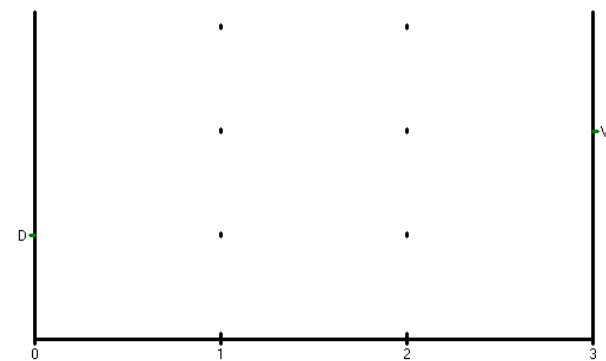
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337L - Rodley – Ring Road Roundabout – Stream 2

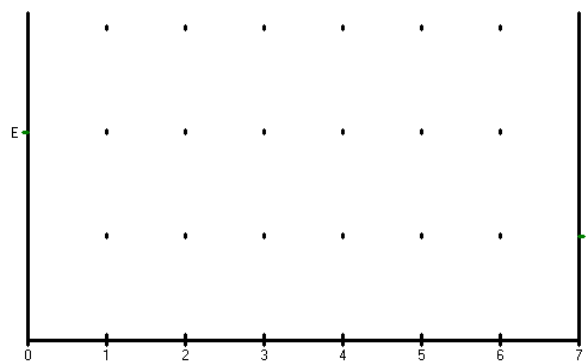
Change From 4 To 5



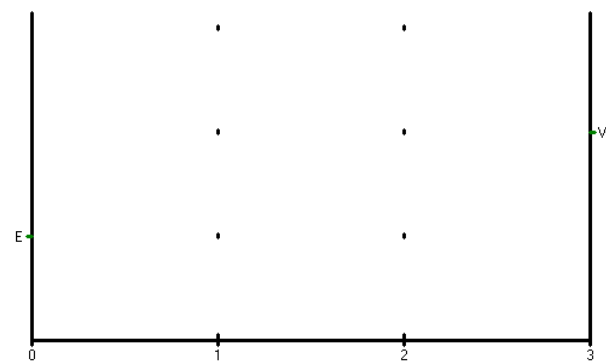
Change From 4 To 6



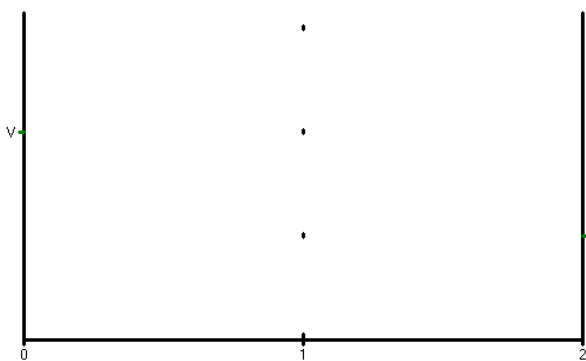
Change From 5 To 4



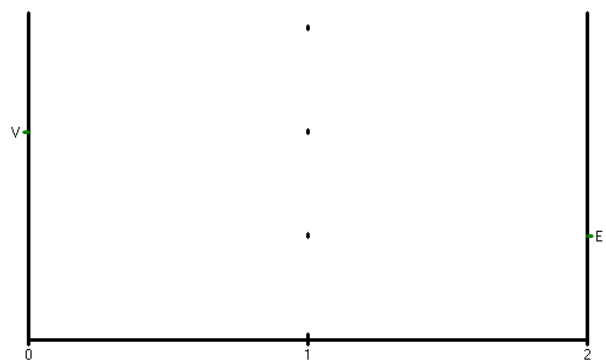
Change From 5 To 6



Change From 6 To 4



Change From 6 To 5



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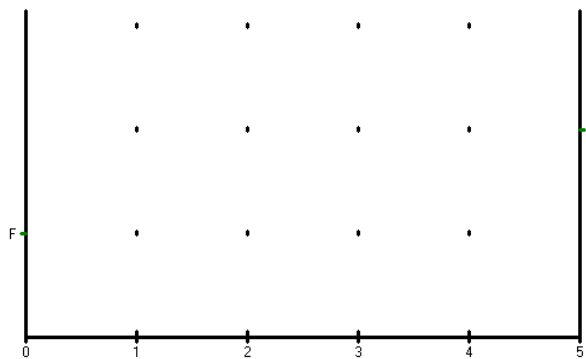
Date:- 15/06/15

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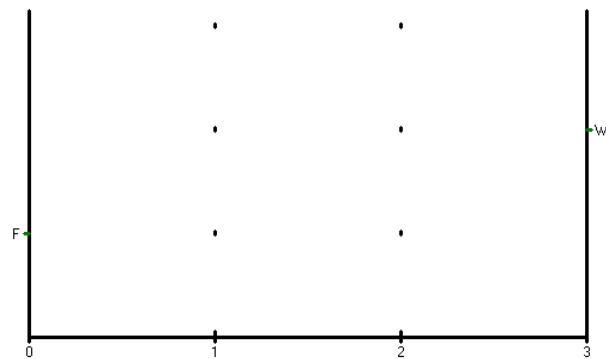
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337L - Rodley – Ring Road Roundabout – Stream 3

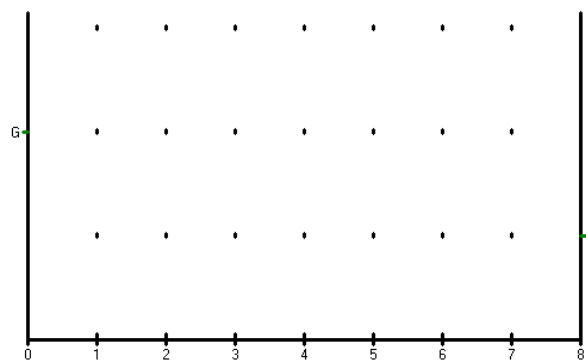
Change From 7 To 8



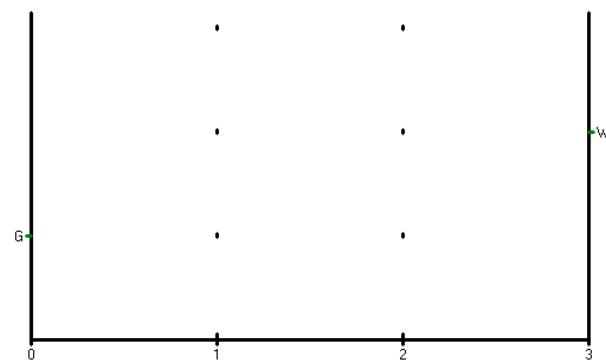
Change From 7 To 9



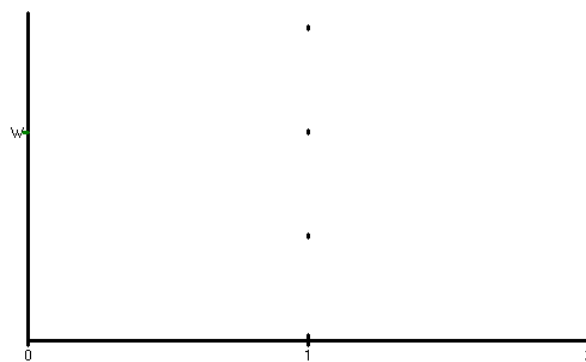
Change From 8 To 7



Change From 8 To 9



Change From 9 To 7



Change From 9 To 8



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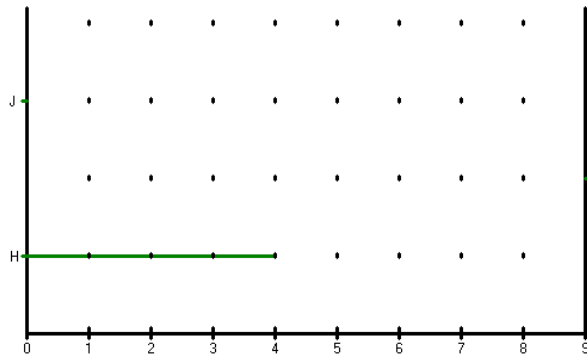
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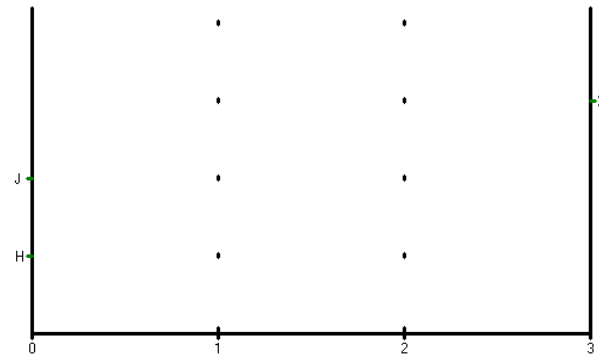
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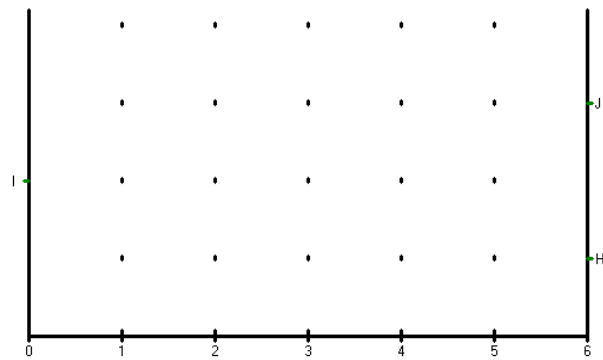
Change From 10 To 11



Change From 10 To 12



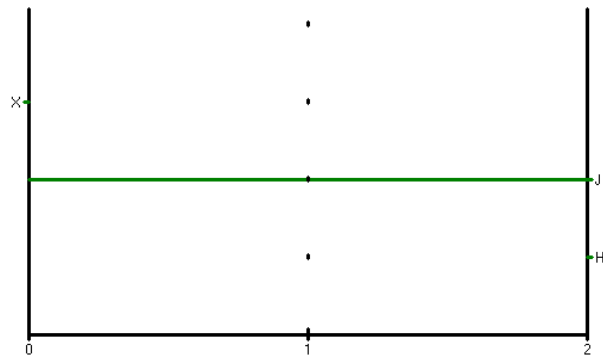
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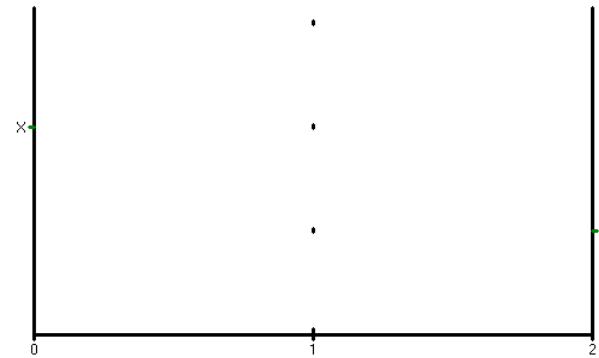
Change From 11 To 12



Change From 12 To 10

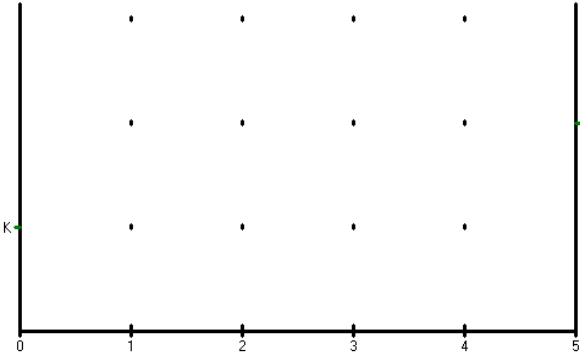


Change From 12 To 11



337L - Rodley – Ring Road Roundabout – Stream 5

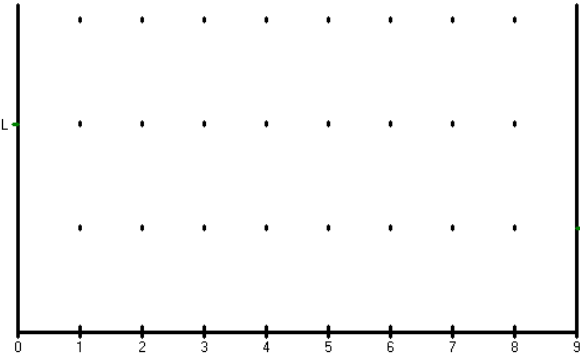
Change From 13 To 14



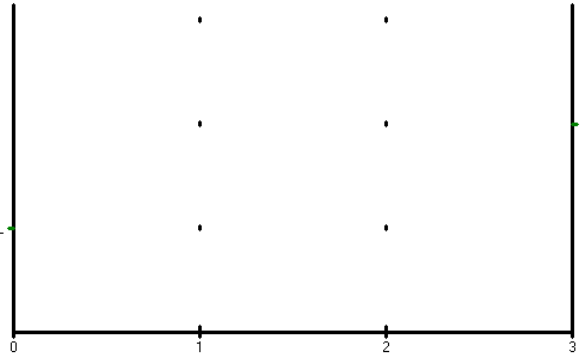
Change From 13 To 15



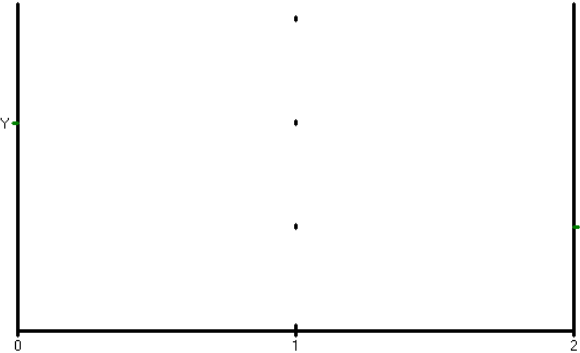
Change From 14 To 13



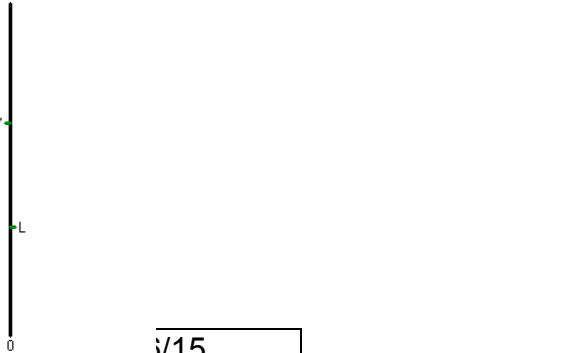
Change From 14 To 15



Change From 15 To 13



Change From 15 To 14



15

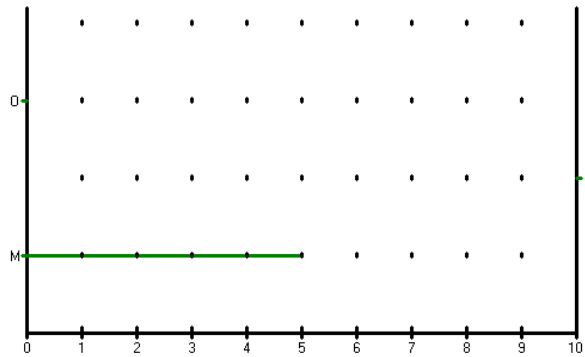
ORN:- 337L

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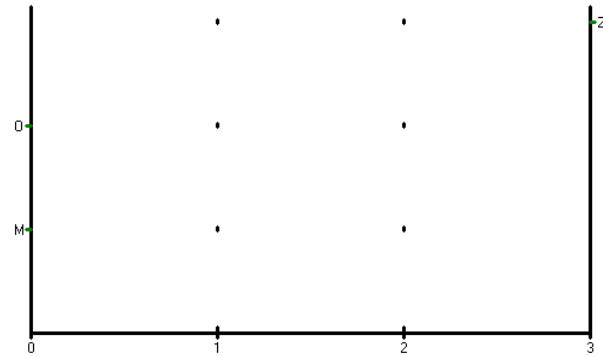
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337L - Rodley – Ring Road Roundabout – Stream 6

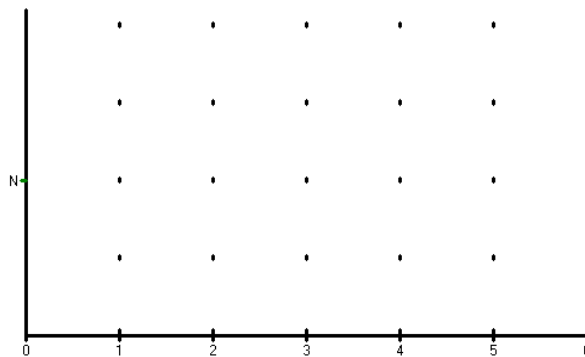
Change From 16 To 17



Change From 16 To 18



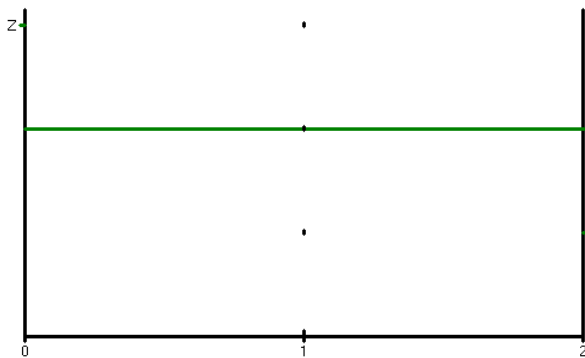
Change From 17 To 16



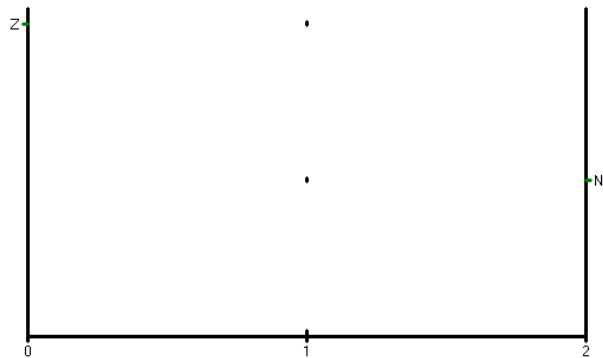
Change From 17 To 18



Change From 18 To 16

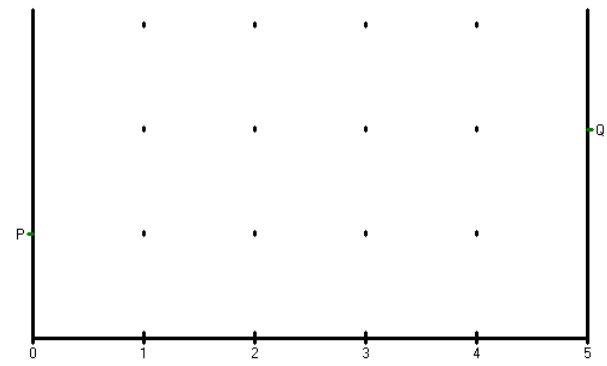


Change From 18 To 17

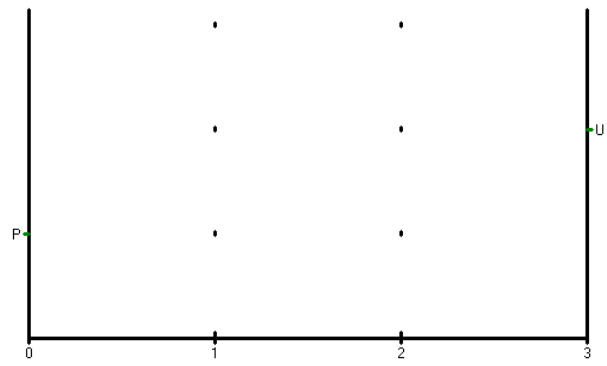


337L - Rodley – Ring Road Roundabout – Stream 7

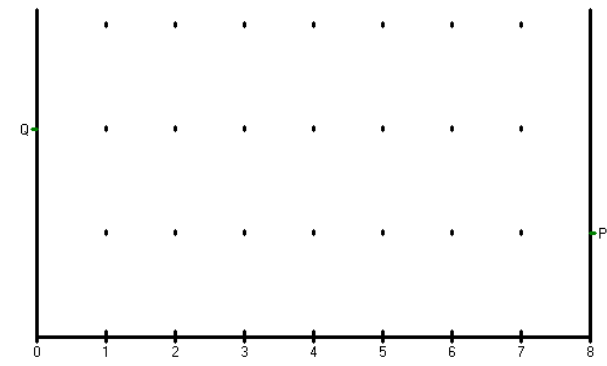
Change from 19 to 20



Change from 19 to 21



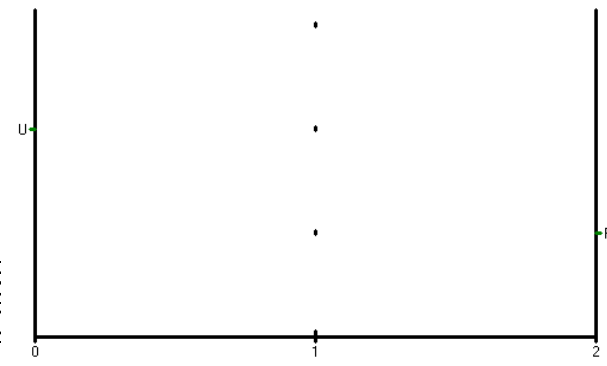
Change from 20 to 19



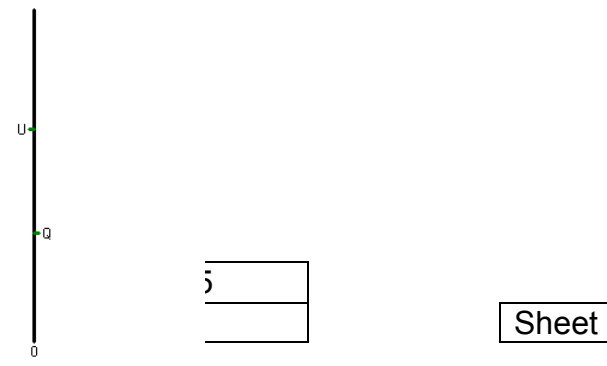
Change from 20 to 21



Change from 21 to 19



Change from 21 to 20

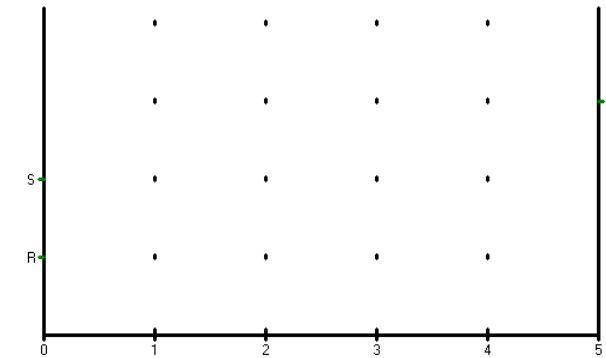


Σ
C

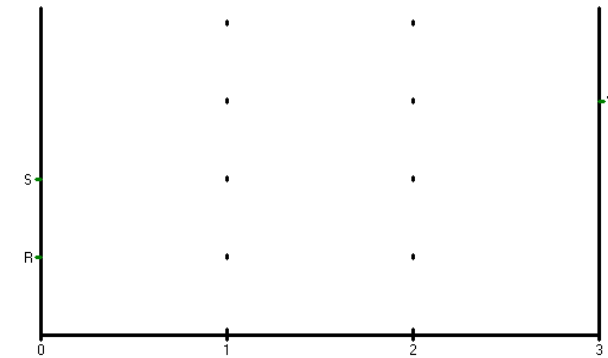
5

337L - Rodley – Ring Road Roundabout – Stream 8

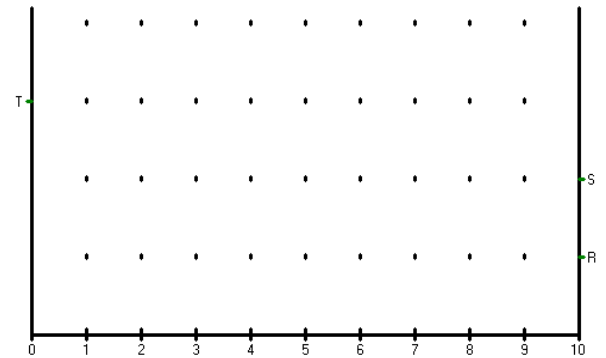
Change from 22 to 23



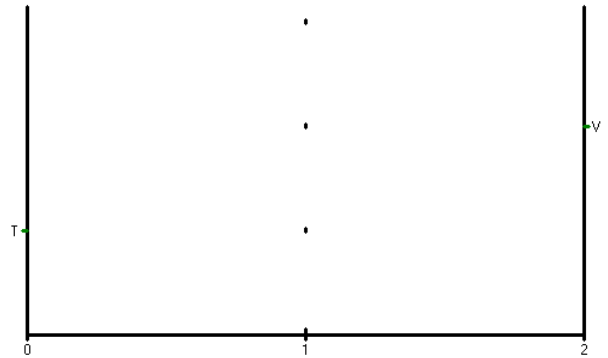
Change from 22 to 24



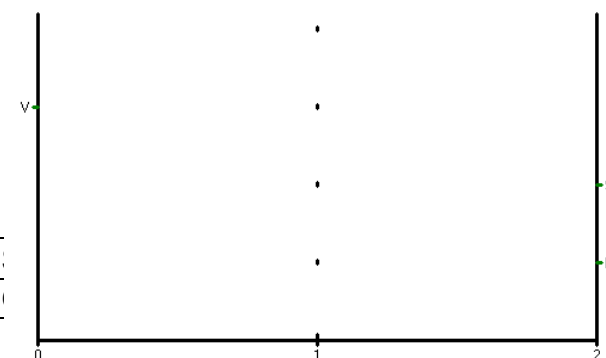
Change from 23 to 22



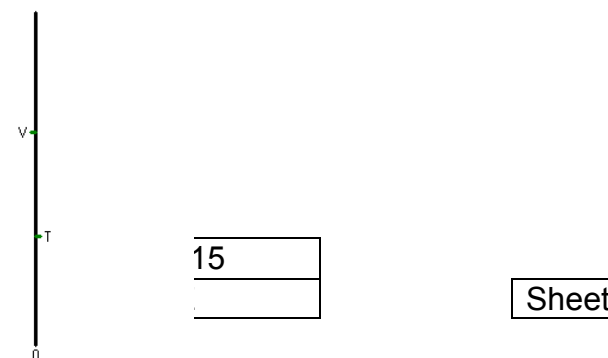
Change from 23 to 24



Change from 24 to 22



Change from 24 to 23



PHASES	PHASE INTERGREEN HANDSET LIMIT VALUES																										MIN GREEN VALUE			
	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X	Y	Z	A1	A2	LOW	HIGH
A		5	.																		3								7	
B	5		5																		3								7	
C	.	6																			3								5	
D					5																	3							7	
E				5																		3							7	
F							5																3						7	
G						6																	3						5	
H									5	.														3					7	
I								5		5														1					7	
J								.	6															3					5	
K												5													0				7	
L											6														3				5	
M													5	.												3			7	
N												5		5												0			7	
O												.	6													3			6	
P																	5										3		7	
Q																6											0		5	
R																			.	5								3	7	
S																			.		5							3	7	
T																		8	8									0	6	
U	2	2	0																										1	
V				2	2																								1	
W						2	0																						1	
X								2	2	0																			1	
Y											2	0																	1	
Z													2	2	0														1	

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where 'N' is number of stage to move to

PROHIBITED / ALTERNATIVE STAGE MOVEMENTS UP TO 16 STAGES

FROM STAGE	TO STAGE																		
	16	17	18	19	20	21	22	23	24										
16		✓	✓																
17	✓		✓																
18	✓	✓																	
19					✓	✓													
20				✓		✓													
21				✓	✓														
22								✓	✓										
23							✓		✓										
24							✓	✓											
								</											

TICK AS REQUIRED		
MODE	These Restrictions are for	No Restrictions
Urban Traffic Control	✓	
Cableless Linking	✓	
Fixed Time	✓	
Hurry Call		
Manual		
Priority		
Emergency		

NOTE:

PERMITTED MOVES

✓ = Permitted

P = Prohibited

X = Ignore

'N' = ALTERNATIVE MOVES

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where 'N' is number of stage to move to

DETECTOR AND PUSH BUTTONS - LOCATIONS AND FUNCTIONS												Detector Types
												Detector Supply Voltage
												DFM Period *Enhanced Y/N *
N°	Location / Title	Det Exn	Phase Dem Ext		Non Latching	Uni Dirctn	Extrnl	Delay Call Cncl		Det DFM(1)	Speed Detn	Special Instructions
1	Ped C Ring Rd Southbnd Enter	.	C	.								Not on DFM
2	Ped G Ring Rd Southbnd Exit	.	G	.								Not on DFM
3	Ped J Ring Rd Northbnd Enter	.	J	.								Not on DFM
4	Ped L Calverley Lane Exit	.	L	.								Not on DFM
5	Ped O Calverley Lane Enter	.	O	.								Not on DFM
6	Ped Q Ring Rd South Enter	.	Q	.								Not on DFM
7	Ped T Rodley Lane	.	T	.								Not on DFM
8												
9												
10												
11												
12												
13												
14												
15												
16												
17												
18												
19												
20												
21												

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22												
23												
24												

Note 1 Y= Use Input on DFM Fail
A= Force Active on DFM Fail
I=Force Inactive on DFM Fail

Note 2 ***For Enhanced DFM Timings Refer To Front Sheet***

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ADDITIONAL SWITCHES & LED'S (ON MANUAL PANEL)		
POSITION	USE	MOMENTARY *
SWITCH 1		
LED 1		
SWITCH 2		
LED 2		
SWITCH 3		
LED 3		
HURRY CALL AUX 4		
HIGHER PRIORITY AUX 5		

* NORMAL ACTION IS A TOGGLE ACTION
(PRESS ONCE FOR ACTIVE, PRESS AGAIN TO MAKE INACTIVE)

PUT ✓ IF MOMENTARY ACTION IS REQUIRED
(ACTIVE ONLY WHILE BUTTON IS PRESSED)

IF SWITCH IS MOMENTARY OR NOT USED THEN
THE ASSOCIATED LED IS AVAILABLE FOR
OTHER PURPOSES

HURRY CALL(S) HC(0) ALWAYS HAS PRIORITY OVER HC(1)					
HURRY CALL N°	CALL STAGE	SECONDS			NOTES
		CALL DELAY	CALL HOLD	CALL PREVENT	
HURRY CALL	LOW				
LIMIT VALUES	HIGH				

OPTION FILTER GREEN ARROW IN MANUAL			
TO APPEAR	NOT TO APPEAR	DEMAND DEPENDANT	INSERT TICK (✓)

PART TIME MODE	
SWITCH OF IN STAGE(S)	

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EXTEND ALL RED FACILITY - (BY DETECTOR) CONSIDERED AS PART OF INTERGREENS

Stage to Stage transition for which ALL RED extension required					
UNIT N°	From stage/ To stage	Intergreen Affected	Ext. Time	All Red Max.	Associated Detector
1					
2					
3					
4					
5					
6					
7					

EXTEND ALL RED TIMINGS				
STREAMS	0	1	2	3
EXTENSION TIME (SECS)*				
ALL RED MAXIMUM TIME (SECS)*				

* MEASURED FROM POINT AT WHICH FIRST PHASE IN NEW STAGE WOULD NORMALLY START ITS RED-GREEN TRANSITION PERIOD.

RED EXTENSIONS	ENTER CHOICE
MANUAL	
CLF	
UTC	
HC	
PRIORITY	
EMERGENCY	
VA	
FT	

For Choice Enter As Follows:-

O = Operative
M = Extend to Maximum
N = No extensions (Not approved for UK)

If table is left blank all modes will be operative

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MODE PRIORITY

ASSIGN NUMBER - 1 = TOP PRIORITY
2 = SECOND PRIORITY etc.
x = NOT USED

____ PART TIME
____ EMERGENCY VEHICLE
____ HURRY CALL(S)
__1__ URBAN TRAFFIC CONTROL
__2__ SELECTED MANUAL CONTROL (1)
____ SELECTED FT or VA or CLF (2)
__3__ CABLELESS LINKING
____ BUS PRIORITY
____ EITHER VEHICLE ACTUATED
__4__ OR FIXED TIME

(1)

MANUAL FACILITY ENABLED	
ALWAYS	✓
WHEN HANDSET PLUGGED IN	
WHEN 'MND' COMMAND ENTERED	

(2)

FACILITY SWITCH POSITION	NOT OPERATIVE INSERT (X)
VA	
FIXED TIME	
CLF	

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PERMANENTLY DISABLED MODES

MODES TO BE DISABLED	STREAM			
	0	1	2	3
PART TIME (3)				
EMERGENCY VEHICLE				
HURRY CALL(S)				
SELECTED MANUAL CONTROL (3)				
URBAN TRAFFIC CONTROL				
SELECTED FIXED TIME OR VA				
CABLELESS LINKING				
BUS PRIORITY				
VEHICLE ACTUATED				
FIXED TIME				
	TICK (✓) IF REQUIRED TO BE DISABLED			

(3) THESE MODES MAY ONLY BE DELETED ON ALL MODES SIMULTANEOUSLY

FIXED CYCLE TIME MODE STAGE SEQUENCE								
---	--	--	--	--	--	--	--	--

STAGE	A	B	C	D	E	F	G	H
Stage	1st	2nd	3rd	4th	5th	6th	7th	8th
Time								

OR

ALTERNATIVELY FIXED CYCLE RUNNING TO CURRENT TIMETABLED MAXIMA																							
--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--	--

Phase	A	B	C	D	E	F	G	H	I	J	K	L	M	N	O	P	Q	R	S	T	U	V	W	X
Demand Dependent																								

MANUAL SELECTION		
------------------	--	--

Button N°	Stage(s) Called	Street Names - If required
0	3,6,9,12, 15,18,21, 24	All Red
1	1,4,7,10, 13,16,19, 22	Effective All Red – internals at green, external approaches at red, exit peds at green
2	-	
3	-	
4	-	
5	-	
6	-	
7	-	

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URBAN TRAFFIC CONTROL																
USE OF BITS											OTU MANUFACTURER AND TYPE:-					
Bit N°	1	2	3	4	5	6	7	8	9	10	11	12	13	14	15	16
Control	SO	F1	F2	F3	F4	F5	F6	F7	F8	F9	D8	F10	F11	F12	F13	F14
Reply	MC	G1	G2	F3	G4	G5	G6	G7	G8	G9	DR8	G10	G11	G12	G13	G14
Secret Sign																Related Phase/Stage
Secret Sign with security																
Notes:-																
0141 CONTROL AND REPLY BIT NOTATIONS																
CONTROL BITS		REPLY BITS									UTC CONTROL REQUIRED WHEN:- (ONE OLY)					
Force (F1,F2,etc) Stage Demand (D1,D2,etc) Common Demand (DX) SW FAC (SF1,SF2) CLF Group Sync (SG) RTC Sync (TS or TS1) Fall Back Mode Select (FM) Lamps On (LO) Local Link Inhibit (LL) Transmission Confirm (TC) Solar Switch Over-ride (SO) Pedestrian Demand (PX) Hold Vehicle Stage (PV) Solar Switch Over-ride (SO)		Stage Confirm (G1,G2,etc) Stage Dem Confirm (DR1,DR2,etc) SW FAC Confirm (SC1,SC2,etc) CLF Group Sync Confirm (CS) CLF Group 0 Confirm (GR1) RTC Sync Confirm (CC or CC1) Fall Back Confirm (FC) Lamps Off Confirm (LE) Hurry Call Confirm (HC) Manual Control Mode (MC) Controller Fault (CF) Detector Fault (DF) Remote Reconnect (RR) One or more Signal Lamps Failed (LF) Pedestrian Inhibit by Lamp Monitor (PI) Pedestrian Reply (GX)									____✓____ Force Bits Present ____ Force Bits and TC/TO Present ____ TC/TO Present B.T. termination requirements are:- _____ B.T. Line Number:- _____					

LEEDS UTC Spec Note : CS Control Bit To Set Internal Clock To 12:00 Midday

OPTION 1 (MCE 0105/0106)											
OPTION 2 (TCD316) (Delete one)											
INSERT PHASES OR STAGES TO BE DEMANDED											
IF DEMANDS TO BE UNLATCHED ADD #											
D8	D14	D20	D23								
G	L	Q	T								
INSERT PHASES TO BE EXTENDED											
D8	D14	D20	D23								
INSERT PHASE OR STAGES FOR REPLY											
D8	D14	D20	D23								
9	15	21	24								
DEMAND DEPENDENT FORCES											
INSERT DEMANDS TO BE CONSIDERED											

CSR Reply Bit To Have 3 Second Duration

D8	D14	D20	D23								
G	L	Q	T								

NOTES:-

- Hash (#) indicates demand dependent force
- Asterisk (*) denotes Control or Reply associated with separate Pelican

URBAN TRAFFIC CONTROL																
USE OF BITS											OTU MANUFACTURER AND TYPE:-					
Bit N°	17	18	19	20	21	22	23	24	25	26	27	28	29	30	31	32
Control	F15	D14	F16	F17	F18	F19	F20	F21	D20	F22	F23	F24	D23	CS		
Reply	G15	DR14	G16	G17	G18	G19	G20	G21	DR20	D22	G23	G24	DR23	CSR	DF	LF1
Secret Sign																Related Phase/Stage
Secret Sign with security																Related Phase/Stage
Notes:-																
0141 CONTROL AND REPLY BIT NOTATIONS																
CONTROL BITS		REPLY BITS					UTC CONTROL REQUIRED WHEN:- (ONE OLY)									
Force (F1,F2,etc) Stage Demand (D1,D2,etc) Common Demand (DX) SW FAC (SF1,SF2) CLF Group Sync (SG)		Stage Confirm (G1,G2,etc) Stage Dem Confirm (DR1,DR2,etc) SW FAC Confirm (SC1,SC2,etc) CLF Group Sync Confirm (CS) CLF Group 0 Confirm (GR1) RTC Sync Confirm (CC or CC1) Fall Back Confirm (FC) Lamps Off Confirm (LE)					☒ Force Bits Present ☐ Force Bits and TC/TO Present ☐ TC/TO Present B.T. termination requirements are:- _____ B.T. Line Number:- _____									
RTC Sync (TS or TS1) Fall Back Mode Select (FM) Lamps On (LO) Local Link Inhibit (LL) Transmission Confirm (TC)		Hurry Call Confirm (HC) Manual Control Mode (MC) Controller Fault (CF)														
Solar Switch Over-ride (SO)		Detector Fault (DF) Remote Reconnect (RR) One or more Signal Lamps Failed (LF) Pedestrian Inhibit by Lamp Monitor (PI)														
Pedestrian Demand (PX) Hold Vehicle Stage (PV) Solar Switch Over-ride (SO)		Pedestrian Reply (GX)														

**LEEDS UTC Spec Note : CS Control Bit To Set Internal Clock To 12:00 Midday
CSR Reply Bit To Have 3 Second Duration**

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URBAN TRAFFIC CONTROL																	
USE OF BITS												OTU MANUFACTURER AND TYPE:-					
Bit N°	33	34	35	36	37	38	39	40	41	42	43	44	45	46	47	48	
Control																	
Reply	LF2																
Secret Sign																	Related Phase/Stage
Secret Sign with security																	Related Phase/Stage
Notes:-																	
0141 CONTROL AND REPLY BIT NOTATIONS																	
CONTROL BITS				REPLY BITS				UTC CONTROL REQUIRED WHEN:- (ONE OLY)									
Force (F1,F2,etc) Stage Demand (D1,D2,etc) Common Demand (DX) SW FAC (SF1,SF2) CLF Group Sync (SG) RTC Sync (TS or TS1) Fall Back Mode Select (FM) Lamps On (LO) Local Link Inhibit (LL) Transmission Confirm (TC) Solar Switch Over-ride (SO) Pedestrian Demand (PX) Hold Vehicle Stage (PV) Solar Switch Over-ride (SO)				Stage Confirm (G1,G2,etc) Stage Dem Confirm (DR1,DR2,etc) SW FAC Confirm (SC1,SC2,etc) CLF Group Sync Confirm (CS) CLF Group 0 Confirm (GR1) RTC Sync Confirm (CC or CC1) Fall Back Confirm (FC) Lamps Off Confirm (LE) Hurry Call Confirm (HC) Manual Control Mode (MC) Controller Fault (CF) Detector Fault (DF) Remote Reconnect (RR) One or more Signal Lamps Failed (LF) Pedestrian Inhibit by Lamp Monitor (PI) Pedestrian Reply (GX)				_____✓_____ Force Bits Present _____ Force Bits and TC/TO Present _____ TC/TO Present B.T. termination requirements are:- _____ B.T. Line Number:- _____									

LEEDS UTC Spec Note : CS Control Bit To Set Internal Clock To 12:00 Midday

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CSR Reply Bit To Have 3 Second Duration

UTC ONLY - MANUAL MODE, NO LAMP SUPPLY, LAMP FAILURES etc., CONFIRM									
TICK AS APPROPRIATE:-	G1/G2	MC	Remote Reconnect	DF	LF1	LF2			
MANUAL MODE SELECTED AND OPERATIVE	✓	✓							
NO LAMP POWER and/or SIGNALS OFF	✓								
DETECTOR FAULT				✓					
SELECT SWITCH NOT IN NORMAL POSITION		✓							
ANY LAMP FAILURE					✓				
2nd RED LAMP FAILURE ON ANY PHASE					✓	✓			

Leeds UTC Lamp Monitoring Specification:

On any lamp failure the LF1 reply bit shall be returned.

On the 1st red lamp failure to any phase conflicting with a pedestrian phase, the LF1 reply bit shall be returned and the detector fault monitor lamp shall flash.

On any second red lamp failure to any phase LF2 shall be returned.

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On any second red lamp failure to any phase conflicting with a pedestrian phase, the LF2 reply bit shall be sent, the detector fault monitor lamp shall continue to flash, the conflicting pedestrian phase wait indicators shall be illuminated and the conflicting pedestrian phase demands shall be inhibited. All other non-conflicting pedestrian phases shall continue to be serviced.

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Telent Timetable Data

TIMETABLE DATA

№	Day Type	Time	Event List №
1	WKD	10:00	1
2	WKD	07:00	2
3	WKD	15:30	3
4	WKD	19:00	4
5	WND	08:00	1
6	WND	19:00	4
7			
8			
9			
10			
11			
12			
13			
14			
15			
16			

Day Types:-

MON	- Monday	XMO	- All except Mon
TUE	- Tuesday	XTU	- All except Tue
WED	- Wednesday	XWE	- All except Wed
THU	- Thursday	XTH	- All except Thu
FRI	- Friday	XFR	- All except Fri
SAT	- Saturday	XSA	- All except Sat
SUN	- Sunday	XSU	- All except Sun

WKD - Mon - Fri
 WND - Sat - Sun
 WEK - Every day

EVENT LIST DATA

Event List №	Event Action 1		Event Action 2		Event Action 3		Event Action 4	
	Type	Params	Type	Params	Type	Params	Type	Params
1	TCF	1						
2	TCF	2						
3	TCF	3						
4	TCF	4						
5	TCF	OFF						

Event Types:-

TTS - timetable phase timing set
 TCF - timetable CLF
 TEF - timetable event flag
 TPT - timetable part-time
 TSF - timetable switched sign
 TDO - timetable detector override
 TDT - timetable detector timing set

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Telent CLF Data

Plan № - 1 DAY

Delay Time - 0

Cycle time - 60

Group	Stream 1			Stream 2			Stream 3			Stream 4		
№	Offset Time		22	Offset Time		01	Offset Time		33	Offset Time		22
	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage
01	00	IM	1	20	IM	4	00	DD	8	35	IM	10
02	34	IM	2	00	IM	5	01	HS	-	00	IM	11
03							10	IM	7			
04												
05												
06												
07												
08												

Influence may be one of the following:-

- IM - immediate move to stage()
- DM - demand dependant move to stage()
- HS - hold current stage()
- PX - prevent moves except to stage()
- AI - add immediate move to stage() to existing influence
- IS - isolate to vehicle actuation
- D - delete currently displayed group from plan

Plan № - 1 DAY

Delay Time - 0

Cycle time - 60

Group	Stream 5			Stream 6			Stream 7			Stream 8		
№	Offset Time		25	Offset Time		21	Offset Time		11	Offset Time		19
	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage
01	00	DD	14	00	IM	16	00	DD	20	00	DD	23
02	01	HS	-	35	IM	17	01	HS	-	01	HS	-
03	10	IM	13				10	IM	19	10	IM	22
04												
05												
06												
07												
08												

Plan № -

Delay Time -

Cycle time -

Group												
№	Offset Time		0	Offset Time		0	Offset Time		0	Offset Time		0
	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage
01												
02												
03												
04												
05												
06												
07												
08												

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Telent CLF Data

Plan № - 2 AM PEAK

Delay Time - 0

Cycle time - 60

Group	Stream 1			Stream 2			Stream 3			Stream 4		
№	Offset Time		02	Offset Time		05	Offset Time		00	Offset Time		31
	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage
01	00	IM	1	23	IM	4	00	DD	8	20	IM	10
02	25	IM	2	00	IM	5	01	HS	-	00	IM	11
03							10	IM	7			
04												
05												
06												
07												
08												

Influence may be one of the following:-

- IM - immediate move to stage()
- DM - demand dependant move to stage()
- HS - hold current stage()
- PX - prevent moves except to stage()
- AI - add immediate move to stage() to existing influence
- IS - isolate to vehicle actuation
- D - delete currently displayed group from plan

Plan № - 2 AM PEAK

Delay Time - 0

Cycle time - 60

Group	Stream 5			Stream 6			Stream 7			Stream 8		
№	Offset Time		34	Offset Time		03	Offset Time		00	Offset Time		34
	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage
01	00	DD	14	25	IM	16	10	IM	19	00	DD	23
02	01	HS	-	00	IM	17	59	DD	20	01	HS	-
03	10	IM	13				00	HS	-	10	IM	22
04												
05												
06												
07												
08												

Plan № -

Delay Time -

Cycle time -

Group												
№	Offset Time		0	Offset Time		0	Offset Time		0	Offset Time		0
	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage
01												
02												
03												
04												
05												
06												
07												
08												

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Telent CLF Data

Plan № - 3 PM PEAK

Delay Time - 0

Cycle time - 60

Group	Stream 1			Stream 2			Stream 3			Stream 4		
№	Offset Time		27	Offset Time		29	Offset Time		36	Offset Time		17
	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage
01	00	IM	1	24	IM	4	00	DD	8	00	IM	10
02	25	IM	2	00	IM	5	1	HS	-	40	IM	11
03							10	IM	7			
04												
05												
06												
07												
08												

Influence may be one of the following:-

- IM - immediate move to stage()
- DM - demand dependant move to stage()
- HS - hold current stage()
- PX - prevent moves except to stage()
- AI - add immediate move to stage() to existing influence
- IS - isolate to vehicle actuation
- D - delete currently displayed group from plan

Plan № - 3 PM PEAK

Delay Time - 0

Cycle time - 60

Group	Stream 5			Stream 6			Stream 7			Stream 8		
№	Offset Time		00	Offset Time		27	Offset Time		26	Offset Time		00
	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage
01	00	DD	14	27	IM	16	00	DD	20	00	IM	22
02	01	HS	-	00	IM	17	1	HS	-	50	DD	23
03	10	IM	13				10	IM	19	51	HS	-
04												
05												
06												
07												
08												

Plan № -

Delay Time -

Cycle time -

Group												
№	Offset Time		0	Offset Time		0	Offset Time		0	Offset Time		0
	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage
01												
02												
03												
04												
05												
06												
07												
08												

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Telent CLF Data

Plan № - 4 NITE

Delay Time - 0

Cycle time - 60

Group	Stream 1			Stream 2			Stream 3			Stream 4		
№	Offset Time	22		Offset Time	01		Offset Time	33		Offset Time	22	
	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage
01	00	IM	1	20	IM	4	00	DD	8	35	IM	10
02	34	IM	2	00	IM	5	01	HS	-	00	IM	11
03							10	IM	7			
04												
05												
06												
07												
08												

Influence may be one of the following:-

- IM - immediate move to stage()
- DM - demand dependant move to stage()
- HS - hold current stage()
- PX - prevent moves except to stage()
- AI - add immediate move to stage() to existing influence
- IS - isolate to vehicle actuation
- D - delete currently displayed group from plan

Plan № - 4 NITE

Delay Time - 0

Cycle time - 60

Group	Stream 5			Stream 6			Stream 7			Stream 8		
№	Offset Time	25		Offset Time	21		Offset Time	11		Offset Time	19	
	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage
01	00	DD	14	00	IM	16	00	DD	20	00	DD	23
02	01	HS	-	35	IM	17	01	HS	-	01	HS	-
03	10	IM	13				10	IM	19	10	IM	22
04												
05												
06												
07												
08												

Plan № -

Delay Time -

Cycle time -

Group												
№	Offset Time	0		Offset Time	0		Offset Time	0		Offset Time	0	
	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage	Start Time	Inf.	Stage
01												
02												
03												
04												
05												
06												
07												
08												

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SPECIAL CONDITIONING - GENERAL REQUIREMENTS

Mode during which conditions apply :- URBAN TRAFFIC CONTROL / CABLESS LINKING

	Stream 1
1 – 2	$\bar{F}1 \bullet F2$
1 – 3	$\bar{F}1 \bullet F3$
2 – 3	$\bar{F}2 \bullet F3$
2 – 1	$\bar{F}2 \bullet F1$
3 – 1	$\bar{F}3 \bullet F1$
3 – 2	$\bar{F}3 \bullet F2$
	Stream 2
4 – 5	$\bar{F}4 \bullet F5$
4 – 6	$\bar{F}4 \bullet F6$
5 – 6	$\bar{F}5 \bullet F6$
5 – 4	$\bar{F}5 \bullet F4$
6 – 4	$\bar{F}6 \bullet F4$
6 – 5	$\bar{F}6 \bullet F5$

IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1

D2 - COMPUTER DEMAND BIT FOR STAGE 2

dem C - DEMAND FOR PHASE C

$\overline{\text{ext C}}$ - LACK OF EXTENSIONS FOR PHASE C

max C - MAX TIMER FOR PHASE C EXPIRED

“.” - BOOLEAN “AND”

“+” - BOOLEAN “OR”

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SPECIAL CONDITIONING - GENERAL REQUIREMENTS

Mode during which conditions apply :- URBAN TRAFFIC CONTROL / CABLESS LINKING

	Stream 3
7 – 8	$\bar{F}7 \cdot F8 \cdot (\text{DEM } G + D9)$
7 – 9	$\bar{F}7 \cdot F9$
8 – 9	$\bar{F}8 \cdot F9$
8 – 7	$\bar{F}8 \cdot F7$
9 – 7	$\bar{F}9 \cdot F7$
9 – 8	$\bar{F}9 \cdot F8 \cdot (\text{DEM } G + D9)$
	Stream 4
10 – 11	$\bar{F}10 \cdot F11$
10 – 12	$\bar{F}10 \cdot F12$
11 – 12	$\bar{F}11 \cdot F12$
11 – 10	$\bar{F}11 \cdot F10$
12 – 10	$\bar{F}12 \cdot F10$
12 – 11	$\bar{F}12 \cdot F11$

IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1

D2 - COMPUTER DEMAND BIT FOR STAGE 2

dem C - DEMAND FOR PHASE C

$\bar{\text{ext C}}$ - LACK OF EXTENSIONS FOR PHASE C

max C - MAX TIMER FOR PHASE C EXPIRED

“.” - BOOLEAN “AND”

“+” - BOOLEAN “OR”

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SPECIAL CONDITIONING - GENERAL REQUIREMENTS

Mode during which conditions apply :- URBAN TRAFFIC CONTROL / CABLESS LINKING

	Stream 5
13 – 14	$\bar{F}13 \bullet F14 \bullet$ (DEM L + D15)
13 – 15	$\bar{F}13 \bullet F15$
14 – 15	$\bar{F}14 \bullet F15$
14 – 13	$\bar{F}14 \bullet F13$
15 – 13	$\bar{F}15 \bullet F13$
15 – 14	$\bar{F}15 \bullet F14 \bullet$ (DEM L + D15)
	Stream 6
16 – 17	$\bar{F}16 \bullet F17$
16 – 18	$\bar{F}16 \bullet F18$
17 – 18	$\bar{F}17 \bullet F18$
17 – 16	$\bar{F}17 \bullet F16$
18 – 16	$\bar{F}18 \bullet F16$
18 – 17	$\bar{F}18 \bullet F17$

IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1

D2 - COMPUTER DEMAND BIT FOR STAGE 2

dem C - DEMAND FOR PHASE C

$\overline{\text{ext C}}$ - LACK OF EXTENSIONS FOR PHASE C

max C - MAX TIMER FOR PHASE C EXPIRED

“.” - BOOLEAN “AND”

“+” - BOOLEAN “OR”

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SPECIAL CONDITIONING - GENERAL REQUIREMENTS

Mode during which conditions apply :- URBAN TRAFFIC CONTROL / CABLESS LINKING

	Stream 7
19 – 20	$\bar{F}19 \bullet F20 \bullet (DEM\ Q + D21)$
19 – 21	$\bar{F}19 \bullet F21$
20 – 21	$\bar{F}20 \bullet F21$
20 – 19	$\bar{F}20 \bullet F19$
21 – 19	$\bar{F}21 \bullet F19$
21 – 20	$\bar{F}21 \bullet F20 \bullet (DEM\ Q + D21)$
	Stream 8
22 – 23	$\bar{F}22 \bullet F23 \bullet (DEM\ T + D24)$
22 – 24	$\bar{F}22 \bullet F24$
23 – 24	$\bar{F}23 \bullet F24$
23 – 22	$\bar{F}23 \bullet F22$
24 – 22	$\bar{F}24 \bullet F22$
24 – 23	$\bar{F}24 \bullet F23 \bullet (DEM\ T + D24)$

IF BOOLEAN EXPRESSIONS ARE USED THEN THESE ARE EXAMPLES OF SYMBOLS THAT COULD BE USED:-

F1 - COMPUTER FORCE BIT FOR STAGE 1

D2 - COMPUTER DEMAND BIT FOR STAGE 2

dem C - DEMAND FOR PHASE C

$\bar{\text{ext C}}$ - LACK OF EXTENSIONS FOR PHASE C

max C - MAX TIMER FOR PHASE C EXPIRED

“.” - BOOLEAN “AND”

“+” - BOOLEAN “OR”

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